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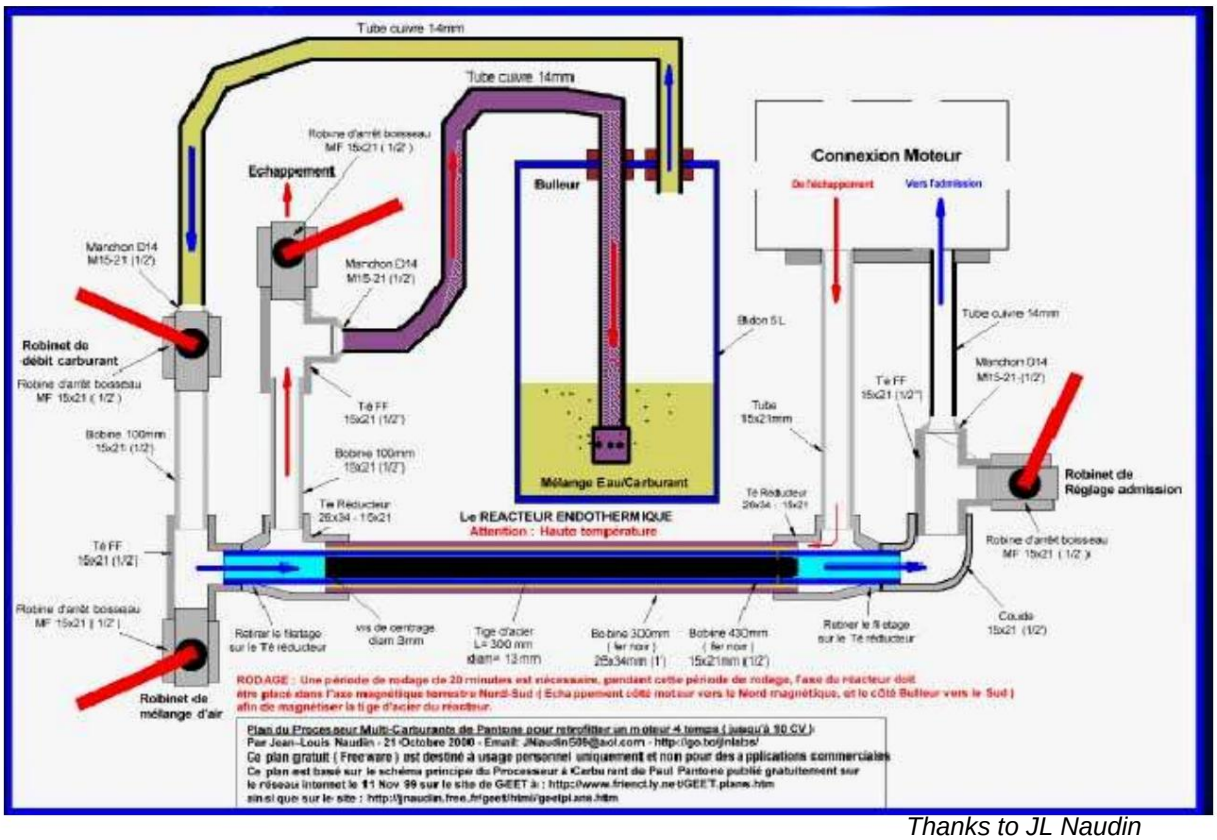
Ion separation in a moving fluid

Hypothesis formulated from the study of the Pantone reactor

Mark C, 09/16/2006

1. Introduction



Thanks to JL Naudin

Admittedly, the Pantone reactor phenomenon is irritating to scientists: how can a simple plumbing assembly improve the performance of a heat engine? In addition, we are dealing with a kind of "black box" equipped with solid padlocks: if we want to place measuring devices, the spell is broken... It is therefore easier for the scientist to cry superstition than to search.

In a first document, I proposed a plausible interpretation of the origin of the relatively high magnetic field present in the core of the reactor (the famous “shuttle” or “rod”). In my mind, the magnetism was just a consequence of operation, but with no real effect on the motor.

Now I am bold enough to propose a model leading to the dissociation of some of the molecules present in the Pantone reactor. Can we affirm that the cracking of water can occur in this way? I'll let you think for yourselves...

Moreover, this model can be transposed to VSLA, and by extension to any system comprising moving ions… Planet Earth included!

Finally, I pushed the study to “magnetic fuel savers”, as well as fluorescent tubes, which I believe are based on a similar principle.

Good reading !

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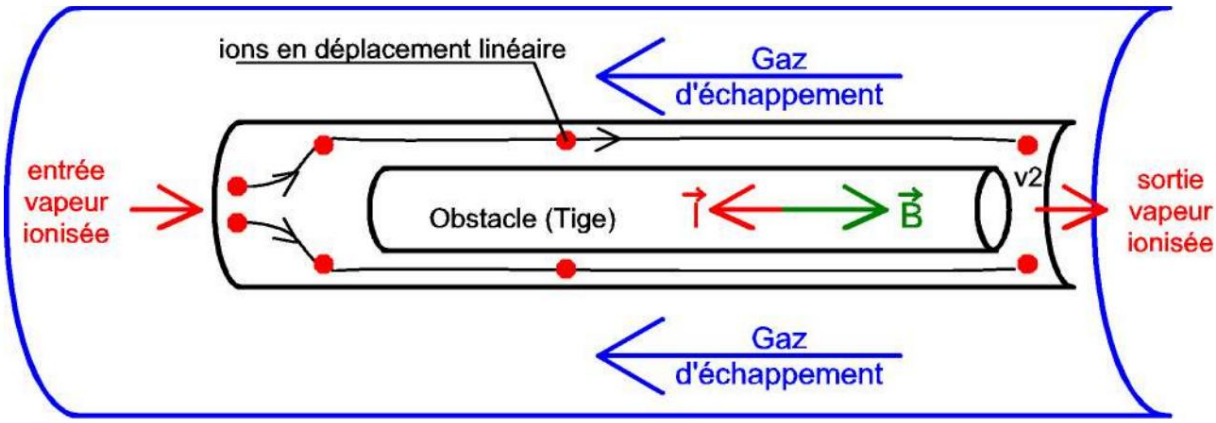
2. Laminar flow (erroneous) assumption

I should have started with this chapter the last time...

When I became interested in the Pantone system (about a year ago), I initially assumed that the molecules of air, water, fuel (except for the “G” system)…Move at a constant rate , therefore accelerated while approaching the rod, but that they maintained a linear trajectory along this rod. Not having any serious knowledge of fluid mechanics, I had no idea what turbulent flow meant.



In this hypothesis, the circulation of ionized molecules, of assumed negative charge (for convenience) creates a resulting electric current, parallel to the field lines of the reactor.
The vectors I (current) and B (induction) are therefore collinear (zero vector product) and therefore do not produce any Laplace force (see later). Moreover, there is no explanation for the mysterious magnetization of the rod, which has given rise to much ridicule from skeptics... However, some report having seen objects "stuck" on reactors in operation (click here and read the § “ C) Analogy with lightning ”)…

However, a minimum of reflection and common sense leads to the conclusion that in this reactor, the life of the ionized molecules, sucked in and heated "white-hot", is far from being a long calm river... But I would not like to plagiarize MM . David and Soarès - whom I warmly thank by the way! Without their explanations and the valuable contributions of the Quanthomme site, I would not have been able to produce this summary.

3. (Incomplete) hypothesis of turbulent flow Then one day… I discovered

the now famous vortex (cf. page devoted to experiments WITH), as well as explanations on the electrification of water vapour.
Suddenly I came to imagine a turbulent flow of ions, which then generates a magnetic field (see previous document).

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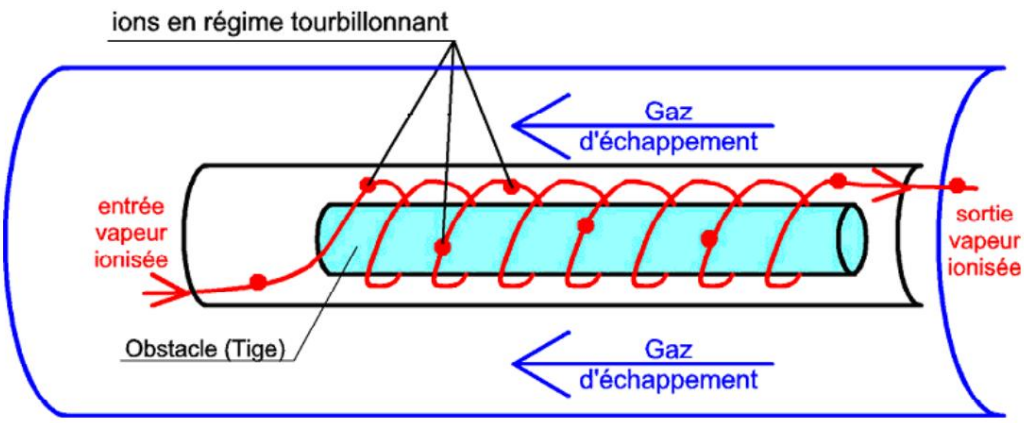

In this hypothesis, negative ions revolve around the nucleus, keeping their distance from each other by mutual repulsion (all of the same sign), and the displacement of the total charge becomes equivalent to an electric current flowing in a conductor. The rod becomes the core of a solenoid (or electromagnet).

A calculation - certainly approximate - made it possible to obtain an induction of almost 0.5 Tesla in a soft iron core, with only a tenth of a microgram of ionized water vapor, rotating 50 times around a 20 cm rod , before leaving the reactor at 200 km/h.

4. What about positive ions?

But the above hypothesis only satisfied me in part because it was based on a crude subterfuge: I had mentally “filtered” the positive ions to keep only the negative ones… Not a very scientific approach. Shame on me !

And this is how remorse tormented me, so much so that I have to publish a denial:

- In fact, the reactor adapts very well to positive ions, it makes no discrimination !

- Better, it's probably the key to its operation!

- And no offense to detractors, Mr. Pantone's suggestion to "prowl" the reactor in the north-south axis is far from being stupid! It's time to rehabilitate this genius *(damn, I'm going to make some enemies)...*

5. To keep it simple

Ultimately, in my opinion, the reactor uses for its own benefit the MHD dear to Mr. Jean Pierre Petit, so simple and yet so little known!

It is the transposition to fluids of Laplace's law, using the famous “3 fingers rule” dear to electrical engineers.

A conductor element dl traversed by a current I, moving in an induction magnetic field B, undergoes a force dF such that:

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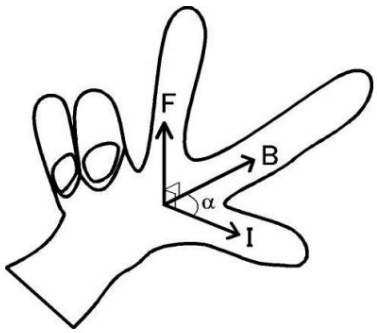


$$d\vec{F} = I\,d\vec{l} \wedge \vec{B}$$

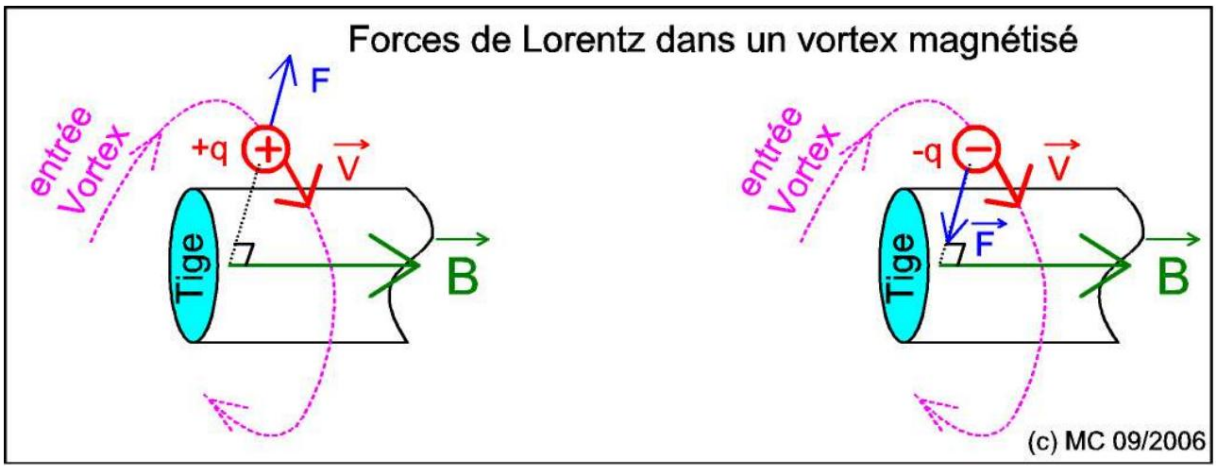
The force will be maximum if dl and B are at right angles (ÿ=90°). This is the principle used in most electric motors.

Now let's replace the conductor and its current with an ion that moves around the rod.

In this case, it is more convenient to replace Laplace's law with its lesser-known counterpart in electrostatics, Lorentz's law :

$$\vec{F} = q\vec{V} \wedge \vec{B}$$

Where the current is replaced by a charge particle q, driven by a speed V:



A moving positive ion (left) generates the equivalent of a positive current, moving tangentially to the rod. The direction of this current follows an angle close to 90° with respect to the induction vector B. It is therefore subjected to a force which tends to move it away from the rod.

This is called the electrodynamic effect: in an electromagnet, the winding tends to move away from the core under the effect of the induction which generated it. This winding can even burst, if the wire is too thin with regard to the force involved!

…and since, of course, no electrician is stupid enough to let his transformer “explode”, we forget that the driver would like to run away. The ion can do it without difficulty. I concealed this fact in my previous bulletin.

But where it becomes amusing… It is to replace a positive ion, by its negative counterpart, which traverses the same trajectory: because the equivalent current becomes negative.

Well, it's surprising: if we compare the two scenarios, it seems that the reactor does not manage the ions in the same way? We therefore come to an essential conclusion:

In this configuration, the positive ions will be "ejected" against the wall of the reactor, while the negative ions will rub against the inner rod...

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Naturally it suffices to reverse the direction of the vortex or the orientation of the rod, to have an opposite result.

6. My theory on the principle of the reactor

In principle, the Pantone reactor aspires:

- hydrocarbons (except our famous "G" system),

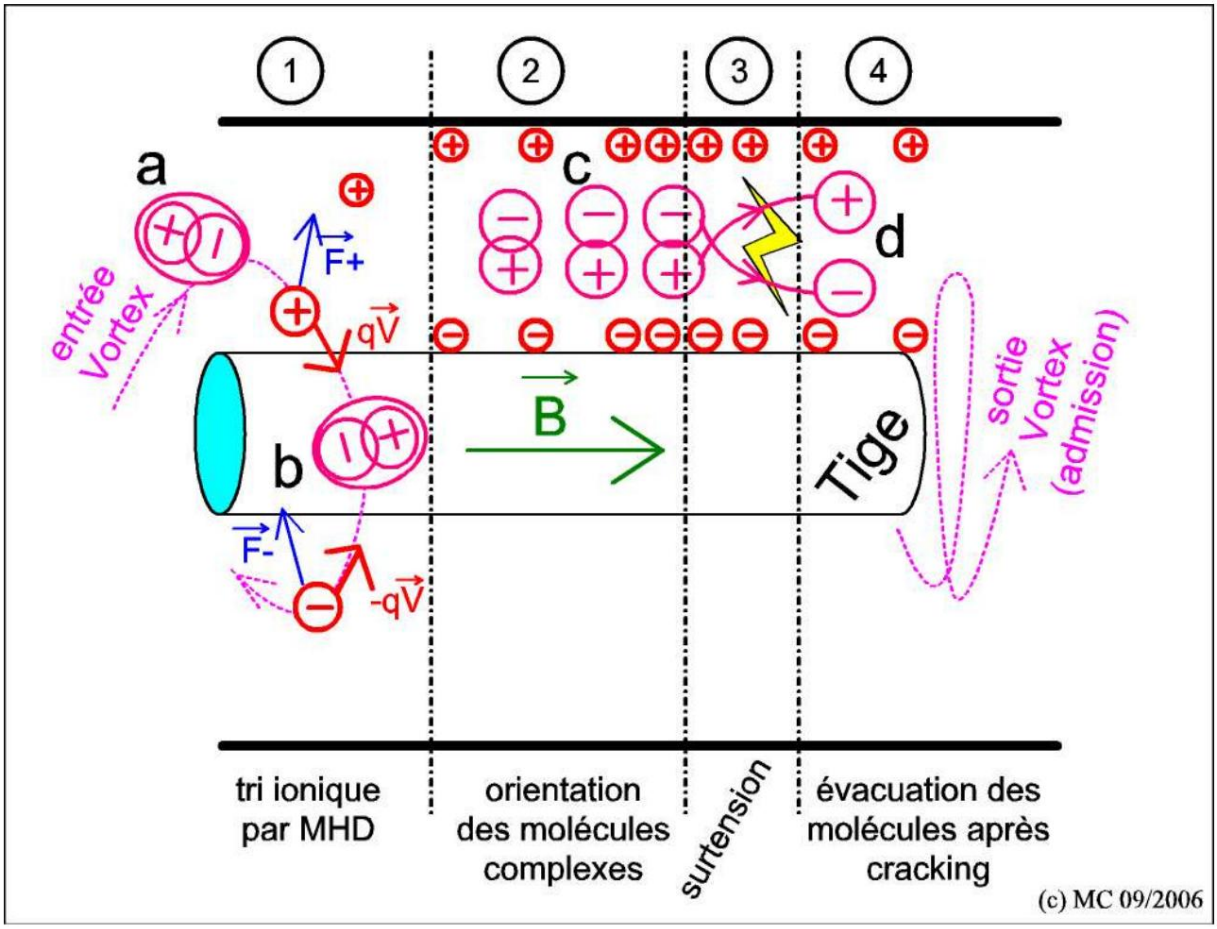
- air (more or less stale alas, but that is not the question…),

- water vapour.

A part of the molecules is stable, another is already ionized, another is made up of negative and positive ions “stuck” together by electrostatic attraction: hydrocarbons for example?

Represent:

- In red, the ions present in the mixture sucked into the reactor inlet (probably, among others, hydroxide and hydronium ions, see on this subject the document dedicated to “ the electrification of water vapour ”).
- In pink, complex molecules (hydrocarbon, …water?), “globally neutral”, but presenting a “dipole moment” allowing them to orient themselves in an electric field, and likely to “break” into two smaller ionized molecules.



This diagram may seem complicated to decipher: let's detail it step by step.

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In the reactor core (exhaust tube not shown) the "work" is done in four phases:

(1) The ions are divided into two layers: some are pressed against the tube, the others against the rod. And it spins, and it rubs, and it heats up (via the exhaust)! On the other hand, the complex molecules (a) and (b) let themselves be blissfully carried in the vortex, simply turning their positive (or negative) side towards the nearest negative (or positive) ion.

It should be noted that at start-up, it is the ***remanent*** *induction of the reactor which decides the tube/ rod distribution.*

Therefore, we understand the famous recommendation of Paul Pantone who had him treated as a charlatan: ***"align the reactor in the North-South axis*** *". This makes it easier to "start" the process by imposing an induction at the start. Especially if the core is not magnetic...*

(2) The electrification is in full swing, the inner surface of the tube is covered with negative charges, while the rod is more and more positive. This distribution is chosen arbitrarily, it suffices to reverse the direction of the vortex or the remanent induction of the rod to have the reverse. Could we distinguish between "positive rod" reactors and "negative rod" reactors?

The complex molecule (c) is "torn" between the armatures of the "condenser" formed by the rod + tube assembly. The covalent force is barely sufficient to hold the two internal ions together. All it takes is a "flick" to separate them.

(3) The accumulation of charges is such that the dielectric breakdown limit is reached: as soon as a molecule passes through the middle of the field (remember that the atmosphere there is rarefied due to the intake depression), it promotes then the conduction of electrons: there is initiation of an electric arc between the rod and the tube. The local temperature easily exceeds 3000°C.

- Some claim to have measured a rod temperature higher than that of the fumes exhaust. This hypothesis goes in this direction.
- By the way, what temperature does it take to "crack" a water molecule under a depression of 800 mbar?

(4) The complex molecule (c), which arrived at (d), found itself split into two ions during the spark. But why these "crossed" trajectories on the diagram? Quite simply because these new ions are in turn subject to Laplace's law, and will rub "like the others" on the overheated metal. Thus the metallic surfaces become electrified again…until the next electric arc.

In a traditional electrolysis there is indeed a "splitting" of the water molecule between the two electrodes, but we do not find this "tilting" phenomenon: on the contrary the ions join the electrodes, hence the appearance of an effect joule as well as a parasitic phenomenon of polarization (fuel cell) which heavily penalizes the energy balance (~50%). The Pantone reactor optimizes the dissociation, its yield is therefore superior to that of conventional electrolysis.

Needless to say that the ions which have undergone such entrainment must find the conditions which will be made for them very mild, once they have arrived in a cylinder of the engine. But this is another story…

7. Additional remarks

Skeptics will say that I overstated the efficiency of the process, because the electric arc mentioned in (3) cannot occur. And this for two main reasons:

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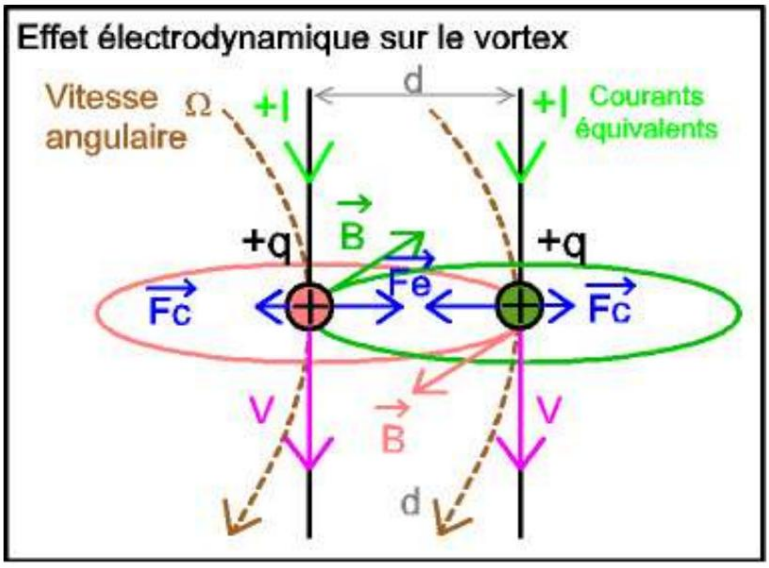
- on the one hand the rod and the tube are generally connected to the same potential. Indeed, the electrons can "bypass" the "Lorentz barrier" by following the metallic contact points between rod and tube: I would indeed recommend the use of an insulated rod !
- the charge density at the surface of the tube and the rod remains low, since the ions of the same sign tend to repel each other.

  This argument is false: the ions rotating in parallel are mutually attracted by electrodynamic effect, each ion bathing in the flux emitted by its neighbour:



The Coulomb force (Fc) is inversely proportional to the distance d between the "turns" of the vortex. It guarantees a “safety distance” between these coils.

But it is supplanted by the electrodynamic force (Fe) which increases with the speed V, itself dependent on the speed of rotation ÿ. The tightening of the turns is therefore done automatically until Fe=Fc.

When I formulated my initial hypothesis of a swirling flow (see §3), I sensed that the ions followed a helical trajectory imposed "by construction", without really being able to define it...

After a few days of reflection, the pieces of the puzzle fell into place more clearly. The following model should explain things better:

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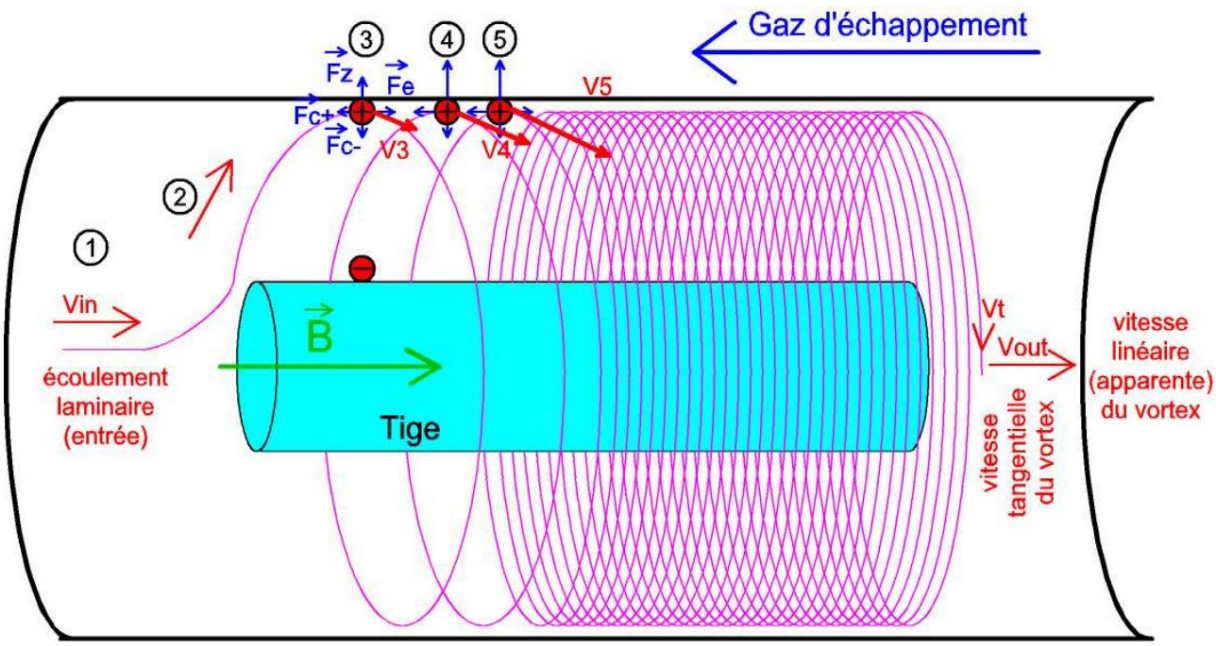

(1) reactor inlet: there is laminar flow at the speed Vin, probably not very different from the “linear” or “apparent” outlet speed Vout. It depends essentially on the depression of the engine.

(2) the ions enter the field of the rod. The positive ions are "plated" on the wall of the tube by the Lorentz force.

(3) A positive ion is subjected to 4 forces which tend to inflect its trajectory:

Fz: Lorentz force. It is she who keeps the positive and negative ions at a distance. It ensures their friction against the metal (tube or rod), improving heat exchange.

Fc-: Coulomb force. Antagonist to the Lorentz force above, but very weak due to the distance between the (+) and (-) ions. Negligible.

Fc+: Coulomb force tending to move it away from ions of the same sign (4) The "coils" cannot touch each other!

Fe: electrodynamic force (see previous illustration). It increases with the speed of the ions. And that's good, because their speed only increases.

Indeed, let us remember that the ions (+) rub against the overheated wall of the tube, by communicating to them a thermal energy.

According to the laws of thermodynamics, this thermal energy communicates a random movement to the ions, causing them to collide with neighboring molecules in a disorderly fashion, thereby increasing the pressure inside the tube (principle of entropy).

I've always wondered why the reactor was operating in depression when, according to the heat balance, the pressure should have increased considerably! But there is hardly any increase in pressure: otherwise the pressure would flow back into the bubbler and nothing would work!

The cause: the ions cannot escape "in disorder" because they are indeed firmly held "on rails" due to the 4 forces seen above.

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Also, the ions have no choice but to accelerate in the already existing direction of rotation, thus converting this thermal energy into speed, and incidentally… into kinetic energy!. In doing so, they take heat from the exhaust gases, which cool them down.

Here is the secret of the effectiveness of the Pantone Reactor:

Almost 100% conversion of
thermal energy into mechanical energy!

If Carnot saw that!

It should be remembered that this energy is also taken from the exhaust gases, considered as a loss by the engine manufacturers. They only use it to "burn off the unburnt" in the catalytic converter. Finally, this pot is useless with a Pantone reactor which rejects few pollutants.

The losses in a traditional mechanical system being irreducible, it is generally considered that the efficiency of a device increases with its size. This small assembly of scrap metal shows us that it is high time to revise our judgement: the mediocre performance of the gigantic EDF turbines is capped at 50%. Half of the energy produced by a thermal power station thus goes “in smoke” (or in clouds for a nuclear power station)! What a shame in the 21st century...

And that's not all: Remember that this "cloud of ions" swirling around the rod, represents the equivalent of a coil traversed by a current. However, this current does not cause a Joule effect, because it does not circulate in a conductor! Dear physicists, Paul Pantone has offered you an ideal lab instrument:

An “infinitely fine” wire coil, with
“adjoining turns”, and…
…an induction without joule losses!

Based on this principle, it should be possible to produce high-power, small-size magnets, without resorting to costly superconductor technology!

Remember that the technique of magnetization by steam has been known since the 19th century, but has so far remained at the stage of laboratory curiosity, for lack of explanation...

8. Generalization

8.1.Ions in turbulent flow

Where it becomes "crispy" is that this model applies to any system equipped with ions "rotating" around an axis. The tourbillon resulting from the AVEC equipping your vehicle, as well as the atmosphere of our good old Earth!

Thus, most vehicle air ducts are made of flexible or rigid plastic, a material well known for its electrostatic properties. The simple circulation of air at high speed in a filter and hoses is enough to charge it – slightly – with ions.

According to fluid mechanics, the ions were, before the initiation of the vortex, in laminar flow and therefore in possession of a kinetic energy function of their mass ($mV^2/2$). HAS

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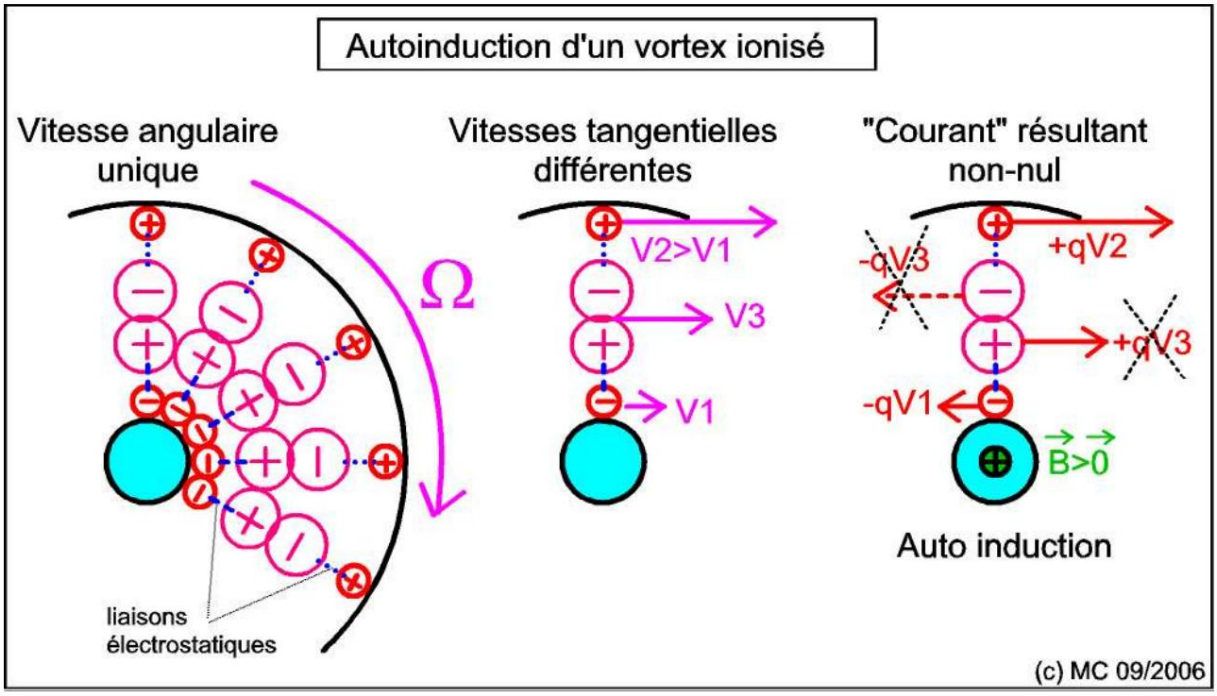
equal mass, the ions closer to the rod should acquire a higher velocity. However, it is not ! This is the paradox observed in several vortex assemblies. Excerpt for example from the document on the “Ranque effect ”:

> Normally, the principle of conservation of angular momentum would require the rotational speed of the internal flux to be greater than that of the external flux so that momentum is conserved. Since these velocities remain equal, part of the energy specific to the initial vortex has gone elsewhere. According to the technicians of the Vortex Corporation, this energy is found in the form of heat in the first part of the jet which leaves the tube. And at the same time, the vortex which sets off again in the opposite direction is cooled in proportions which can be considerable: everything depends on the internal profile of the tube. Such, then, is the explanation provided by the American engineers, but they themselves recognize that it is not sufficient to give a complete interpretation of all the phenomena involved.

With the model used here, we conceive that the positive and negative ions rotate simultaneously since they are "coupled" through complex molecules "globally neutral", but provided with a "dipole moment", and which maintain an electrostatic bond between the rod and the tube. The set of molecules thus set in rotation, finally moves like a nut that goes around its screw!



Ions and molecules certainly all have the same angular velocity, but those located towards the center (here "-") have a tangential velocity V1 lower than that, V2, of the ions located at the periphery (here "+"). The charges of the intermediate half-molecules, which are too close, see their respective influences cancel each other out since they are subjected to the same speed V3.

In the end, we see that, even considering a statistical balance of the number of positive and negative ions present in the vortex, the difference in speed of the positive (V2) and negative (V1) charges generates the equivalent of a current electric (here positive) which would traverse a whorl around the core… The heart of the vortex thus constitutes the core of this coil, and is the seat of an induction… Induction which reinforces the initial magnetic field.

It should be noted that, to "initiate" the ionized vortex, a very weak magnetic field is sufficient, which will then be reinforced by the separation and renewal of the ions.

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An unbalanced distribution of ions at the origin will give the same result without initial induction. However, in an assembly WITH, it is at the periphery of the vortex that the molecules can lose or gain electrons by rubbing against the pipes. A load imbalance is therefore present by construction.

1.1.Examples of application of the theory

Earth's magnetic field

It is attributed to the existence of the Earth's core alone, but researchers are struggling to come up with a viable model.

Instead of looking in the ground, how about looking in the sky? the rotation of the upper layers of the atmosphere could generate this field: at the equator, the tangential speed of the ionosphere, located more than 7000 km from the axis of the Earth, is all the same close to 1800 km/h!

Note that the moon, devoid of atmosphere, does not have an internal magnetic field... One more point to add to this theory.

“Dual temperature plasmas”

This is illustrated in a fluorescent tube: the electrons circulating in the center of the tube are subjected to high temperatures (5000°C is a common value) while the positive ions, located at the periphery of the tube, remain "cold ".

The explanations given on the operation of this commonly used luminaire are not totally satisfactory: we speak evasively of “confinement”, of “absence of collisions” due to the “relative vacuum”. The fact remains that these electrons seem to outrageously violate the laws of physics: they should inevitably join the positive ions if they were subjected to the Coulomb force alone! The enigma is solved if we consider the spontaneous appearance (?) of a vortex which will ensure “ionic partitioning” by self-induction…

8.3. Fluid in laminar flow

More and more vehicle users are resorting to “magic” fuel savers that clip onto the petrol or diesel pipes, unfortunately for a price that is often prohibitive. For homes, it's been a long time since we've been offered anti-scaling systems to be fixed on water pipes.

Here again, our well-meaning censors trust the great specialists, who tell them: “Holy tricks! A magnet has no influence on fuel or limescale! ". Even…

Unlike the Pantone assembly or the WITH, where the prior existence of ions is assumed, I had to accept the idea that complex molecules can themselves be broken down into ions under the effect of Lorentz forces. Now that you know their existence, I let you guess what will happen to a complex molecule, walking in a straight line in a magnetic field perpendicular to the movement:

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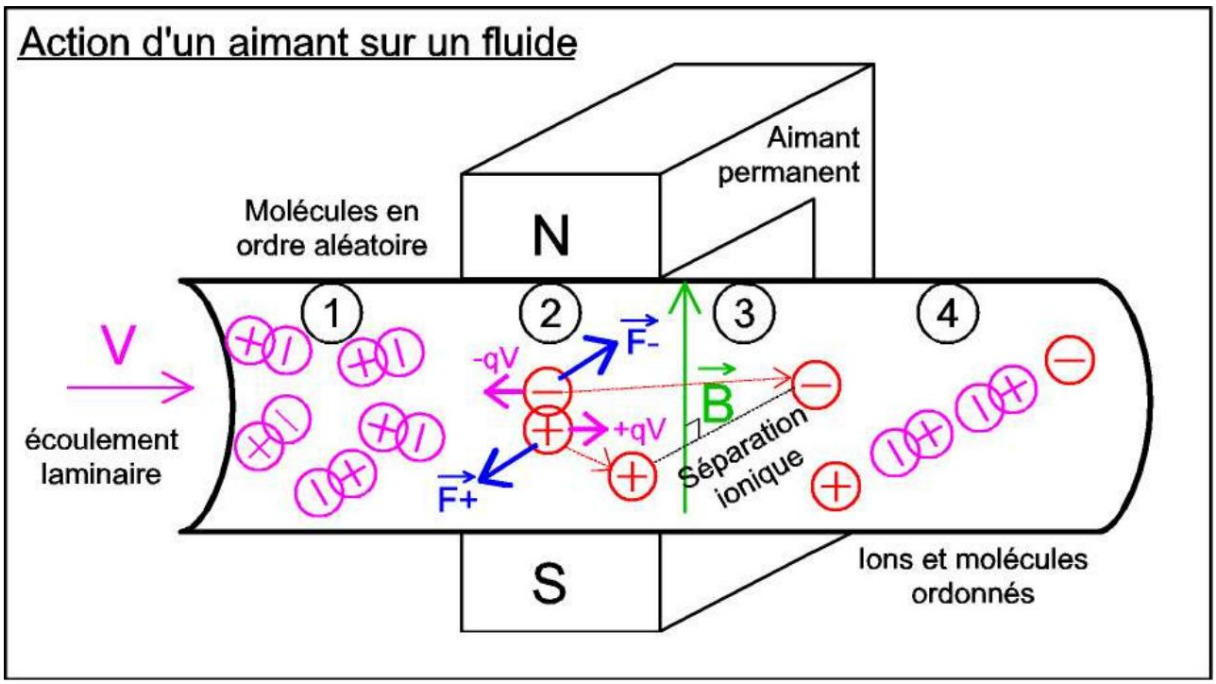
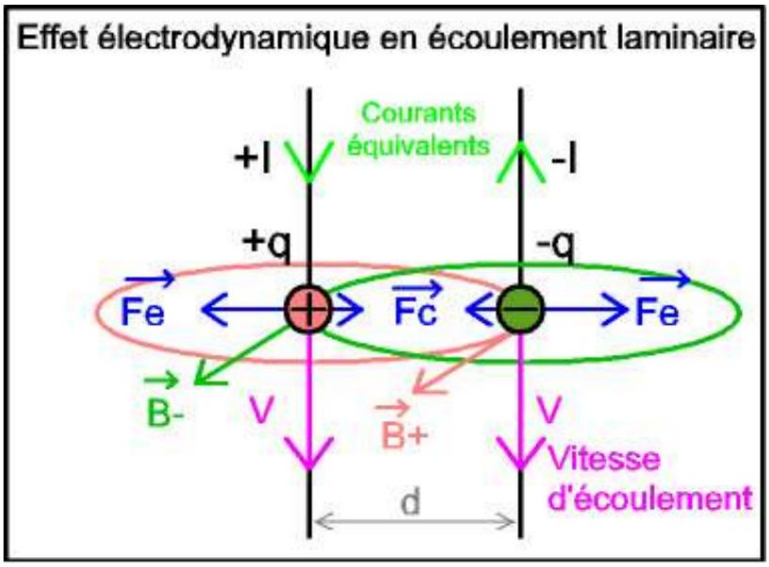


In (1), the complex molecules let themselves be carried around at the same linear speed V, in joyous disorder.

In (2), each molecule is subject to Lorentz forces, which orient this molecule perpendicular to the flow and tend to break the covalent bond of the two ions which compose it.

In (3), separate ions are "plated" on opposite walls of the pipe by the magnetic field.

Even when they leave the magnetic field, they remain distant by electrodynamic effect: indeed each ion moves in a magnetic field emitted by the opposite ion moving at the same speed:



In our hypothesis, the distance d is the section of the pipe, the ions (in red) being “pressed against the walls” under the joint action of the magnet and the flow (see the previous illustration).

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The electrodynamic force of repulsion Fe is generated by the circulation of the ion in the magnetic field generated by its antagonist. It is inversely proportional to the distance d (section of the pipe) and increases with the speed V.

If the speed is sufficient, this force Fe remains greater than the electrostatic force of attraction Fc (Coulomb force), which is independent of the speed and inversely proportional to the square of d.

In (4): the ions being very far apart, the force of attraction between them remains weak. Recombination of these ions is unlikely. In addition, the neighboring molecules, comparable to dipoles, align their faces of opposite sign facing the ions and maintain a certain balance. It can last "a certain time" even in the absence of flow (valve closed).

Naturally, the initial disorder will gradually resume its rights. But the effect must last long enough, since owners of water heaters have noticed the absence of scale in their hot water pipes. Let us remember that each time the user opens the tap, the electrodynamic forces again move away the ions which would have come closer...

9. Conclusion

"Physics is definitely constituted in its fundamental concepts; all it can now provide is the precise determination of a few additional decimal places".

Lord Kelvin, pillar of thermodynamics (1892)

I wonder why no one had yet formulated this kind of hypothesis, yet within the reach of a first-grade student. Perhaps because too many believed that "it was impossible" and were afraid of wasting time by seriously studying the problem?

Or quite simply because the MHD is a field today fallow. Now considered obsolete, it was only studied in the past (until the 1980s) in the hope of producing silent electric thrusters for boats or planes. Under these conditions, only the laminar flow was studied : the distribution of the positive and negative ions by the Lorentz force is of course known, but it was then considered as a simple consequence of the circulation of a current in the fluid. Moreover, one of the advantages of the MHD in propulsion is precisely to eliminate turbulence, and this whatever the speed!

Turbulent phenomena were therefore purely and simply ignored.

I also think that this theory can bring elements of response to certain "inconsistencies" noted experimentally in fluids in circulation, but for which nobody could formulate an explanation: fluctuations of pressure, temperature, flow…

This study is still incomplete. I admit that it is sorely lacking in numerical values.
Everyone has to make up their own mind. And it is up to the experimenters to modify their Pantone reactor or their VSLA if necessary to confirm or invalidate this theory...

It is by getting together that we can hope to progress.

Happy crafting everyone!

Mark C.